

Instrument Recognition – Multitrack Datasets

AI in Music, Winter Term 2025/2026

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Exercise 1 [Visualisation, 1 point]

Multitrack data consists not just of the polyphonic mixes of a piece of music, but also contains the individual instrument “tracks”, sometimes also referred to as “stems”. These tracks are monophonic, meaning that each track (usually) only contains the recording of a single instrument. Later, these tracks can be combined into a full music piece by a producer who may edit, mix and master the tracks to create a coherent listening experience.

Download the multitrack (<https://rwth-aachen.sciebo.de/s/6Y288tKRAQm3yCB>) which was taken from the Mixing Secrets dataset [2]. Then, write a function that plots the waveforms of all tracks and allows you to highlight time ranges that contain instrument activation with a different color.

Exercise 2 [Instrument Activity Detection, 8 points]

Use a method inspired by the MedleyDB [1] paper by Bittner et al. to calculate an instrument activation matrix for the multitrack you downloaded in Exercise 1. The method works by applying the following steps to each track:

1. Split the track into half-overlapping frames of size 4096.
2. Apply half-wave rectification.
3. Calculate the energy in each frame.
4. Normalise the energy per track within the range of $[0, 1]$.
5. Apply a low-pass filter to smoothen the energy across frames.
(use `scipy.signal.butter(2, 0.075, "lowpass", analog=False)`)
6. Calculate activation confidences with the following logistic function:

$$C(i, t) = 1 - \frac{1}{1 + \lambda \cdot e^{(H_{i,t} - \theta)}}$$

where $H_{i,t}$ is the energy of track i in frame t , and λ , θ are tunable parameters affecting slope and activation threshold, respectively.

7. Consider frames for which $C(i, t) \geq 0.5$ as active, and all other frames as inactive.

Find good values for θ and λ by experimenting with them until the activations look reasonable to you. (This is an imprecise science, given what you have learned about the ADSR-Envelope.) Use the function you built in Exercise 1 to visualise the sections of each track where activation is detected.

Exercise 3 [Audio Bleed, 1 point]

If you listen to tracks 9, 10 and 11 you might hear that these tracks contain quieter audio from the other instruments. This phenomenon is also known as “bleed”. How could this bleed effect influence the instrument activity detection accuracy?

References

- [1] Rachel M Bittner et al. “Medleydb: A multitrack dataset for annotation-intensive mir research.” In: *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*. 2014, pp. 155–160.
- [2] Siddharth Gururani and Alexander Lerch. “Mixing secrets: a multi-track dataset for instrument recognition in polyphonic music”. In: *Late Breaking Demos of the International Society for Music Information Retrieval Conference (ISMIR)* (2017), pp. 1–2.